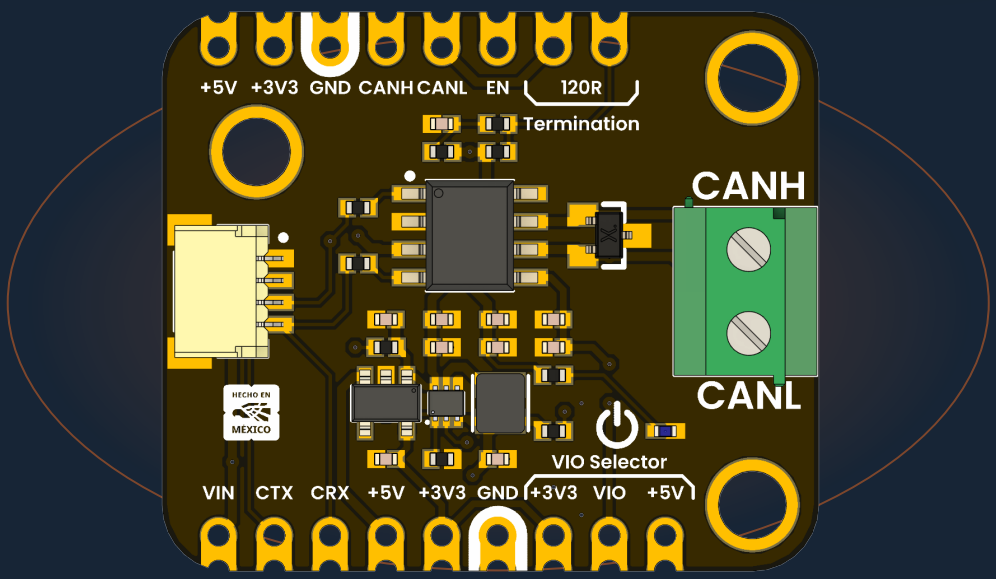
PRODUCT REFERENCE MANUAL

DEVLAB FACTOR

TCAN1051HVD CAN Transceiver Module

Mfr. Part #: UE0085

Version 1.0.0



Fault-protected high-speed CAN physical-layer interface with selectable 3.3 V or 5 V controller logic.

Classic CAN | CAN FD · 2 Mbps | 3.3 V / 5 V I/O | 120 Ω termination

DOCUMENT MAP

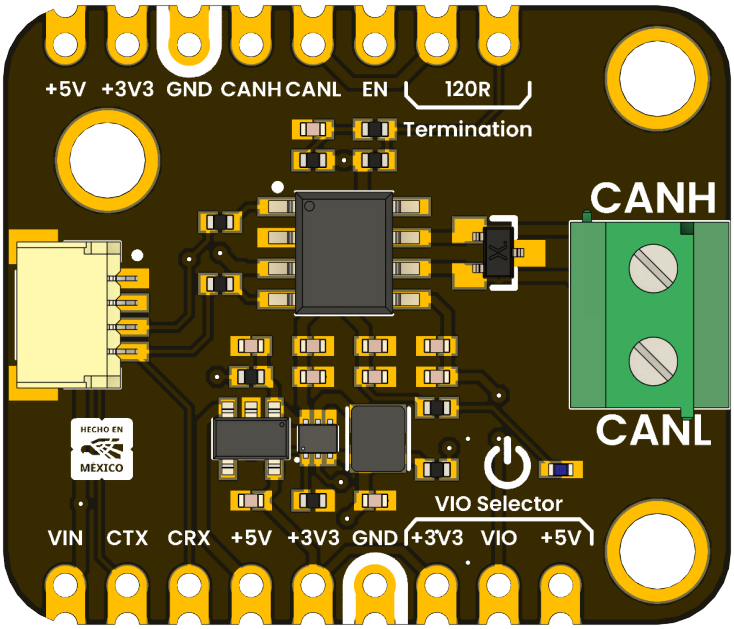
Contents

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Description

The **DevLab TCAN1051HVD CAN Transceiver Module** is a compact physical-layer interface for Classic CAN and CAN FD networks. It connects a microcontroller or stand-alone CAN controller to the differential CAN bus through the CTX (TXD), CRX (RXD), CANH, and CANL signals.

The module is built around the Texas Instruments TCAN1051HVD, a fault-protected high-speed CAN transceiver with a separate VIO supply. An onboard boost converter and 3.3 V regulator generate the transceiver and logic rails from VIN. The VIO selector allows direct use with either 3.3 V or 5 V controllers.



> This module implements the CAN physical layer only. The host must provide a CAN controller or a microcontroller with an integrated CAN/TWAI peripheral.

Applications

- Industrial automation and distributed control
- Robotics and mobile platforms
- Building and environmental control
- Automotive prototyping and diagnostics
- CANopen, DeviceNet, NMEA 2000, ISOBUS, and other CAN-based networks

Hardware Features

- TCAN1051HVD high-speed, fault-protected CAN transceiver
- Classic CAN and CAN FD operation up to 2 Mbps
- Selectable 3.3 V or 5 V logic interface (VIO)
- Onboard TPS61023 5 V boost converter and AP2112K 3.3 V regulator
- Switchable split 120 Ω termination (2 × 60 Ω with 4.7 nF center capacitor)
- PESD2CANFD24L-UX protection device on CANH and CANL
- 22 Ω series resistors on CTX and CRX
- 3.81 mm screw terminal for the CAN bus
- 4-pin, 1 mm-pitch connector for VIN, GND, CTX, and CRX
- Castellated headers for breadboard, carrier-board, or direct-wire integration
- Power indicator LED and exposed regulator enable signal

1 The Board

The DevLab TCAN1051HVD CAN Transceiver Module (UE0085) provides a physical-layer interface between a CAN-capable host controller and a differential CAN bus.

The design separates the controller-side logic interface from the CANH and CANL bus signals. Controller connections are available through a 4-pin JST-compatible connector and castellated contacts, while the CAN bus is accessible through a 3.81 mm screw terminal and dedicated board contacts.

An onboard 5 V boost converter, 3.3 V linear regulator, selectable logic voltage, and CAN bus protection circuitry support integration with 3.3 V and 5 V controllers.

1.1 Accessories

The UE0085 module is supplied with the following accessories:

Accessory	Quantity	Purpose
DevLab TCAN1051HVD CAN Transceiver Module (UE0085)	1	Main CAN physical-layer interface board
JST/Qwiic-compatible cable (AR4561)	1	Connects the module to the host controller through the 4-pin interface
Single-row male header, 1 × 9 pins	1	Provides solderable connections for board integration
Single-row male header, 1 × 8 pins	1	Provides additional solderable connections
Jumper bridge, 2.54 mm	2	Used for VIO logic-voltage selection (3.3 V or 5 V) and enabling the onboard 120 Ω CAN bus termination

The supplied headers and jumper bridges may require installation depending on the intended application.

Select only one VIO voltage level at a time. Enable the 120 Ω termination only when the module is installed at a physical end of the CAN bus.

1.2 Board Identification

Item	Value
Product	DevLab TCAN1051HVD CAN FD Transceiver Breakout
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product type	High-speed CAN / CAN FD physical-layer transceiver module
Main transceiver	Texas Instruments TCAN1051HVD
Power converter	TPS61023
Voltage regulator	AP2112K-3.3
Logic interface	Selectable 3.3 V / 5 V

Item	Value
CAN interface	CANH and CANL differential bus
Manufacturer Part Number (MPN)	UE0085
Available schematic	Revision 1
Product Reference	Version 1.0.0

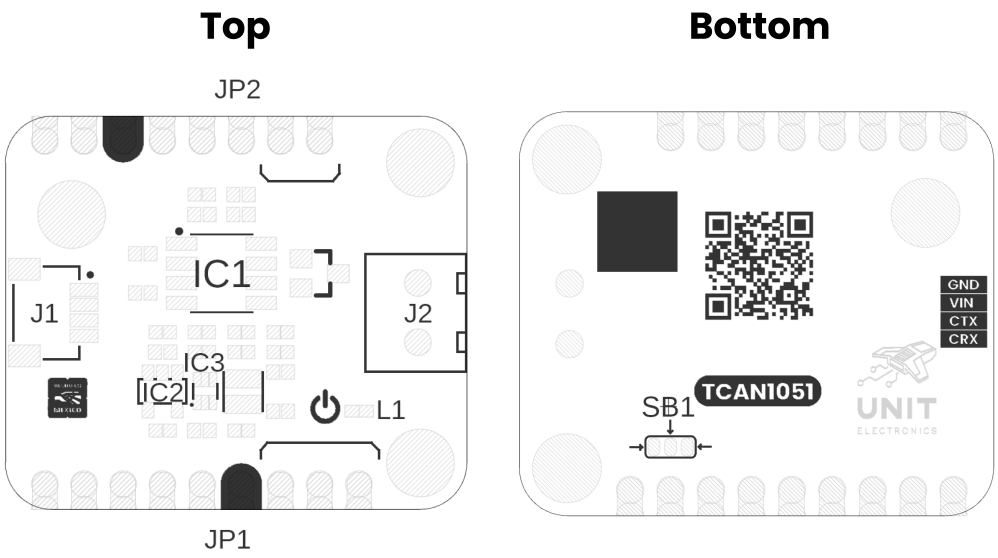
Board, schematic, pinout, and documentation revisions are controlled independently.

1.3 Identified Assemblies

Assembly	Function	Source status
TCAN1051HVD (IC1)	High-speed CAN physical-layer transceiver supporting Classic CAN and CAN FD up to 2 Mbps	Identified in the schematic and product reference
TPS61023 (U1)	Boost converter generating the regulated 5 V rail from VIN	Identified in the schematic
AP2112K-3.3 (U2)	Linear regulator generating the 3.3 V rail from the 5 V supply	Identified in the schematic
PESD2CANFD24L-UX (D1)	Transient protection for CANH and CANL	Identified in the schematic
Green LED (D2)	Indicates the presence of the regulated 5 V supply	Identified in the schematic
Controller connector (J1)	4-pin, 1 mm-pitch connection for VIN, GND, CTX, and CRX	Identified in the schematic and pinout
CAN bus terminal	2-position, 3.81 mm screw terminal for CANH and CANL	Identified in the schematic
Split termination network (R2, R3, C5)	Provides 120 Ω CAN bus termination using two 60 Ω resistors and a 4.7 nF center capacitor	Identified in the schematic
Termination selector (JP1)	Controls the connection of the onboard termination network	Identified in the schematic; assembly configuration requires verification
VIO selector (JP4)	Selects 3.3 V or 5 V transceiver logic levels	Identified in the schematic and pinout
Enable signal (EN)	Controls the TPS61023 boost converter	Identified in the schematic and pinout
Castellated contacts	Provide access to power, control, and CAN bus signals	Identified in the board documentation

The module implements the CAN physical layer only. A host microcontroller with an integrated CAN controller or an external CAN controller is required for communication.

1.4 Board Views



Views of Board Topology

Views of Board Topology

The top view identifies the main integrated circuits, controller and CAN bus connectors, power circuitry, and castellated contacts.

The bottom view shows the VIO solder selector, board identification markings, and controller signal references.

Reference	Description	Location
IC1	TCAN1051HVD CAN transceiver	Top
IC2	AP2112K-3.3 linear regulator	Top
IC3	TPS61023 boost converter	Top
J1	4-pin controller interface connector	Top
J2	2-position CAN bus screw terminal	Top
JP1	CAN bus termination selector	Top
JP2	Controller and power castellated contacts	Top
L1	Boost converter inductor	Top
SB1	VIO solder selector	Bottom
QR code	Board identification and product reference	Bottom
GND, VIN, CTX, CRX	Controller interface signal markings	Bottom

The board topology illustration identifies component placement and interface locations. Electrical connections and component values are defined by the released schematic.

Board artwork and schematic revisions must be verified independently before manufacturing or hardware modification.

1.5 Handling

Use standard electrostatic discharge (ESD) precautions when handling or installing the module.

Remove power before installing headers, changing jumper positions, or modifying the VIO selection.

Select only one VIO voltage level at a time. Connecting the 3.3 V and 5 V rails together may damage the module or the connected host.

Verify CANH and CANL polarity before connecting the module to an active CAN network.

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus. Intermediate nodes must not introduce additional termination resistance.

Do not connect external power directly to the regulated 3V3 or 5V outputs unless the application has been specifically validated for that configuration.

The module does not provide galvanic isolation between the host controller and the CAN bus.

2 Ratings and Electrical Characteristics

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VIN	Module input voltage	3.3	—	5.0	V
VCC	TCAN1051HVD bus supply	4.5	5.0	5.5	V
VIO	Selected I/O supply	2.8	3.3 / 5.0	5.5	V
VOUT(5V)	TPS61023 regulated output, nominal	—	5.0	—	V
VOUT(3V3)	AP2112K regulated output, nominal	—	3.3	—	V
IO(RXD)	Recommended CRX output-current range	−2	—	2	mA
TA	TCAN1051HVD operating free-air temperature	−55	—	125	°C
CAN FD	Supported data rate for TCAN1051HVD	—	—	2	Mbps

VIN represents the recommended module input voltage. VCC, VIO, I/O current, temperature, and CAN FD values are TCAN1051HVD specifications.

The 5 V and 3.3 V rails are generated by the onboard TPS61023 boost converter and AP2112K-3.3 linear regulator, respectively.

The complete assembled module has not been independently qualified over the full IC temperature range.

2.2 Selected Transceiver Characteristics

Unless otherwise noted, the following characteristics apply to the Texas Instruments TCAN1051HVD transceiver.

Symbol	Description	Min	Typ	Max	Unit
VIH	CTX high-level input voltage	0.7 × VIO	—	—	V
VIL	CTX low-level input voltage	—	—	0.3 × VIO	V
VOH	CRX high-level output voltage at −2 mA	0.8 × VIO	—	—	V
VOL	CRX low-level output voltage at 2 mA	—	—	0.2 × VIO	V
VO(DOM), CANH	Dominant CANH output voltage	2.75	—	4.5	V
VO(DOM), CANL	Dominant CANL output voltage	0.5	—	2.25	V
VOD(DOM)	Dominant differential output, 50–65 Ω load	1.5	—	3.0	V

Symbol	Description	Min	Typ	Max	Unit
VCM	Receiver common-mode range	−30	—	30	V
tPROP(LOOP)	Total loop delay	—	100 / 110	160 / 175	ns

2.3 Power Regulation Characteristics

The module integrates a TPS61023 boost converter and an AP2112K-3.3 low-dropout linear regulator.

TPS61023 — 5 V Boost Converter

Symbol	Description	Min	Typ	Max	Unit
VIN(IC)	Converter operating input voltage	0.5	—	5.5	V
VSTART	Converter startup input voltage	—	—	1.8	V
VOUT	Adjustable output voltage range	2.2	—	5.5	V
VOUT(MOD)	Configured module output voltage	—	5.0	—	V
fSW	Switching frequency, VIN > 1.5 V	—	1.0	—	MHz
ILIM	Valley switch current limit, typical	—	1	—	A

The TPS61023 generates the regulated 5 V rail from VIN.

The converter current-limit specification refers to the internal switching current and must not be interpreted as the available module output current.

The maximum external load on the 5 V rail has not been independently characterized.

AP2112K-3.3 — 3.3 V Linear Regulator

Symbol	Description	Min	Typ	Max	Unit
VIN(LDO)	Regulator input voltage, nominal	—	5.0	—	V
VOUT	Fixed regulated output voltage, nominal	—	3.3	—	V
IOUT	Rated regulator output-current capability	—	—	600	mA
VDROP	Dropout voltage	TBD	TBD	TBD	mV
TA	Regulator operating temperature	TBD	—	TBD	°C

The AP2112K-3.3 generates the module’s 3.3 V rail from the regulated 5 V supply.

The 600 mA rating represents the regulator IC capability, not the guaranteed external current available from the assembled module.

Available output current depends on regulator thermal dissipation, PCB layout, input voltage, and total module power consumption.

Dropout voltage and operating-temperature limits require confirmation against the exact AP2112K manufacturer datasheet.

2.4 Absolute Maximum Ratings

Absolute maximum ratings are stress limits, not normal operating conditions.

TCAN1051HVD — CAN Transceiver

Symbol	Description	Min	Max	Unit
VCC	TCAN1051HVD 5 V supply	−0.3	7	V
VIO	TCAN1051HVD I/O supply	−0.3	7	V
VBUS	CANH or CANL, H-suffix device	−70	70	V
VDIFF	CANH-to-CANL differential voltage, H-suffix device	−70	70	V
IO(RXD)	CRX output current	−8	8	mA
TJ	IC junction temperature	−55	150	°C
TSTG	IC storage temperature	−65	150	°C

The ±70 V bus-fault rating applies only to the specified CAN bus pins and must not be applied to VIN, CTX, CRX, EN, VIO, 3V3, or 5V.

Power Regulators

The TPS61023 and AP2112K-3.3 have independent absolute maximum ratings defined by their respective manufacturers.

Their absolute maximum input, output, and temperature ratings must not be interpreted as the operating limits of the complete UE0085 module.

2.5 Module Power and Thermal Considerations

The module uses the following power architecture:

- VIN supplies the TPS61023 boost converter.
- TPS61023 generates the regulated 5 V rail.
- AP2112K-3.3 generates the 3.3 V rail from 5 V.
- TCAN1051HVD operates from the regulated 5 V supply.
- VIO selects either the 3.3 V or 5 V logic rail.
- Both regulated rails are accessible through the module headers.

The AP2112K power dissipation can be estimated as:

$P_D \approx (V_{IN(LDO)} - V_{OUT}) \times I_{OUT}$

At a 5 V regulator input and 3.3 V output, the voltage difference is approximately 1.7 V.

The available external current on the 3.3 V and 5 V rails has not been independently qualified.

Do not exceed the thermal, electrical, or current limits of any individual component.

3 Functional Overview

3.1 Signal Path

CTX drives the TCAN1051HVD TXD input. A logic LOW requests a dominant bus state, and a logic HIGH requests a recessive state. The transceiver converts this logic signal into the CANH/CANL differential output.

CRX is driven by the TCAN1051HVD RXD output. CRX is LOW when the receiver detects a dominant bus state and HIGH when it detects a recessive state. The CRX output level follows the selected VIO rail.

The TCAN1051HVD includes dominant-timeout, thermal-shutdown, and undervoltage protection. The bus and logic pins present high impedance when the transceiver is unpowered.

3.2 Power Architecture

VIN feeds the TPS61023 boost converter, which generates the 5 V transceiver supply. The AP2112K then generates the 3.3 V rail. Both regulated rails are exposed on the castellated headers.

The VIO selector connects the transceiver I/O supply to either 3.3 V or 5 V:

Host logic	VIO selector
3.3 V MCU or CAN controller	3V3
5 V MCU or CAN controller	5V

Select exactly one side of the VIO solder selector. Bridging both sides shorts the 3.3 V and 5 V rails and can damage the module or host.

EN controls the TPS61023 regulator. The onboard bias enables the regulator for normal use; driving EN LOW disables the generated supply rails and transceiver.

3.3 Bus Termination

The onboard termination is a split 120 Ω network made from two 60 Ω resistors. Its midpoint is AC-coupled to ground by 4.7 nF to reduce common-mode noise.

Close the **120R Termination** selector only when this module is installed at one physical end of the CAN trunk. A correctly wired point-to-point bus normally has one 120 Ω terminator at each end (approximately 60 Ω measured between CANH and CANL with power removed). Leave the selector open at intermediate nodes.

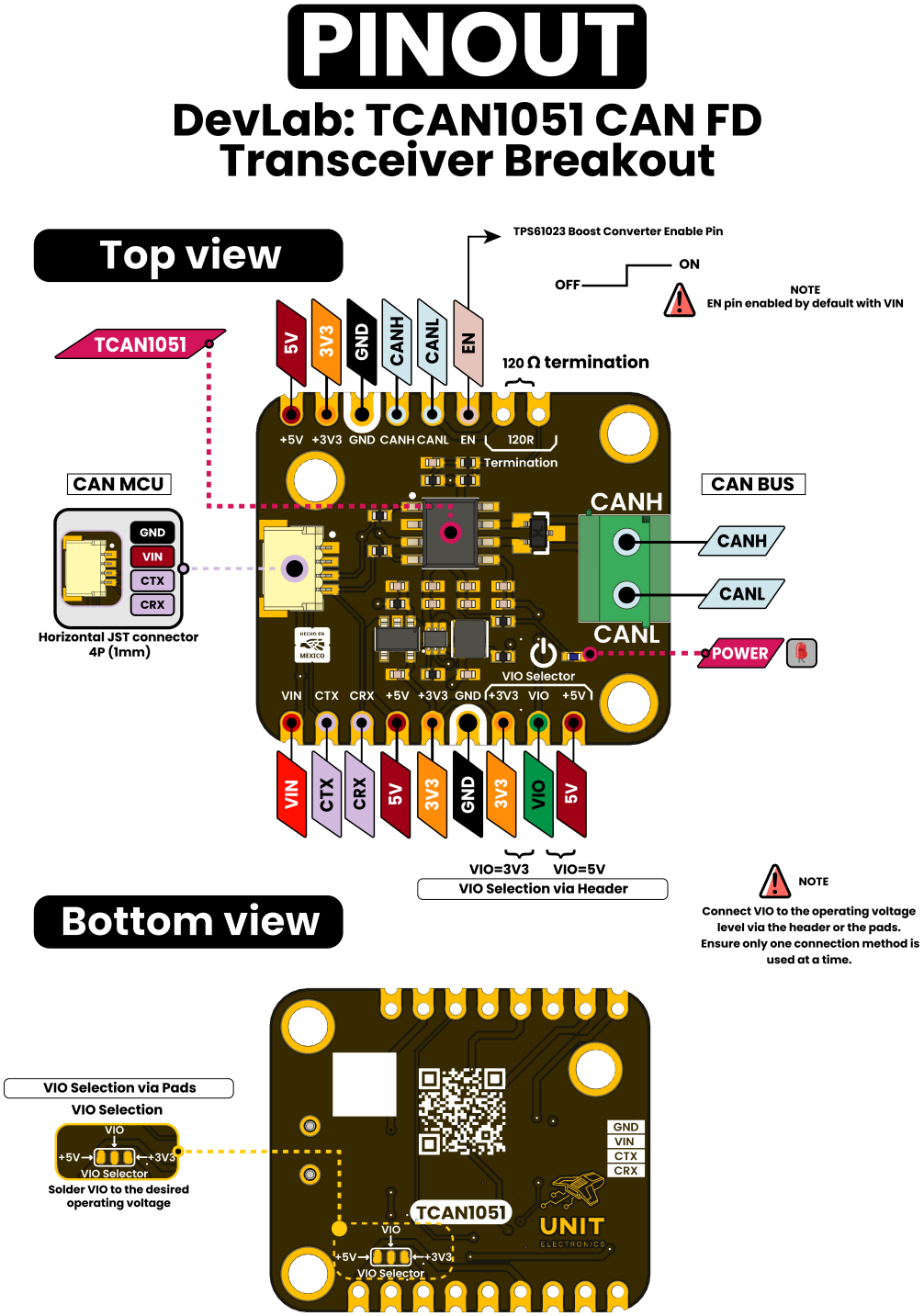
3.4 Bus Protection

CANH and CANL are protected by a PESD2CANFD24L-UX device. The TCAN1051HVD H-suffix transceiver provides a ±70 V bus-fault rating and a ±30 V receiver common-mode range. These protections improve robustness but do not replace correct grounding, cabling, termination, or system-level surge design.

4 Connectors and Pinouts

The UE0085 module provides controller-side logic connections, differential CAN bus terminals, and castellated contacts for integration with external hardware.

The following pinout identifies the available signals, power connections, and onboard configuration options.



4.1 Signal Reference

Label	Direction	Description
VIN	Power input	Module input supply, 3.3 V to 5.0 V
GND	Power	Common module and bus reference
CTX	Input	Controller transmit signal; LOW = dominant
CRX	Output	Controller receive signal; LOW = dominant
CANH	Bus I/O	CAN high bus line
CANL	Bus I/O	CAN low bus line
EN	Input	5 V boost-converter enable; LOW disables the generated supply rails
5V	Power output	Generated 5 V transceiver rail
3V3	Power output	Generated 3.3 V logic rail
VIO	Power	Selected transceiver logic supply (3.3 V or 5 V)

The CTX and CRX signals operate at the voltage level selected through VIO. The host controller must use compatible logic levels.

The module implements the CAN physical layer only. An external CAN controller or a microcontroller with an integrated CAN peripheral is required.

4.2 J1 Controller Connector

J1 is a 4-position, 1 mm-pitch connector providing power and communication signals to the host controller.

Use the bottom-side silkscreen to confirm the pin order before connecting a cable.

Signal	Connection
GND	Host ground
VIN	Host power supply
CTX	Host CAN TX output
CRX	Host CAN RX input

Verify the cable orientation and signal order before connecting the module to an external controller.

4.3 J2 CAN Bus Terminal

J2 is a 2-position, 3.81 mm screw terminal providing the differential CAN bus connections.

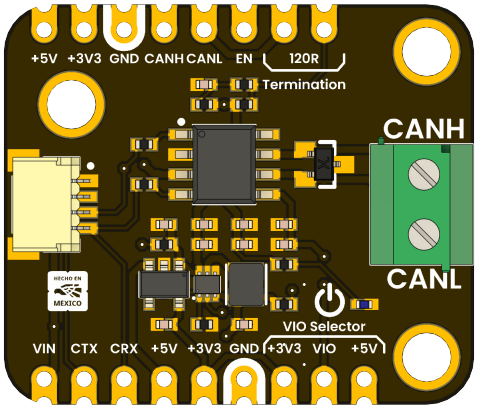
Position	Signal	Connection
Upper	CANH	CAN bus high line
Lower	CANL	CAN bus low line

Use twisted-pair wiring for CANH and CANL. Maintain correct bus polarity and minimize stub lengths to preserve signal integrity.

4.4 Castellated Headers

The castellated contacts provide access to controller signals, CAN bus connections, regulated power rails, and configuration signals.

These contacts support direct wiring, carrier-board integration, and prototyping applications.



The contact labels are printed on the board silkscreen. Refer to the pinout illustration for signal identification.

4.5 Configuration Jumpers

The module includes two jumper-selectable functions: VIO logic-voltage selection and CAN bus termination.

Configuration	Selection	Function
VIO	3V3	Sets the transceiver logic interface to 3.3 V
VIO	5V	Sets the transceiver logic interface to 5 V
CAN termination	Enabled	Connects the onboard 120 Ω termination network
CAN termination	Disabled	Disconnects the selectable termination network

VIO Selection

Select the VIO voltage according to the logic level of the connected host controller.

Only one VIO voltage selection must be active at a time. Connecting both the 3.3 V and 5 V rails together may damage the module or the host controller.

CAN Bus Termination

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus.

For intermediate nodes, leave the termination disabled.

A correctly terminated CAN bus normally uses one 120 Ω termination at each physical end of the main bus.

The termination network and jumper configuration must match the released hardware assembly revision.

5 Board Operation

5.1 Before Connecting the Module

1. Confirm that the host includes a CAN controller or CAN-capable peripheral.
2. With power removed, set VIO to match the host logic voltage (3.3 V or 5 V).
3. Enable the onboard 120 Ω termination only if this node is at an end of the CAN trunk.
4. Check CANH/CANL polarity and provide a common ground where required by the system design.

5.2 Basic Connection

Module	Host or bus
VIN	3.3 V to 5.0 V supply
GND	Host/bus reference ground
CTX	CAN controller TX output
CRX	CAN controller RX input
CANH	CANH trunk conductor
CANL	CANL trunk conductor

Keep the CANH/CANL pair twisted, minimize stubs, and place termination only at the two physical ends of the main bus.

5.3 Power-Up Check

After applying VIN:

- The green power LED should illuminate.
- The 5V rail should measure approximately 5 V.
- The 3V3 rail should measure approximately 3.3 V.
- CRX should idle HIGH at the selected VIO level when the bus is recessive.
- CANH and CANL should both be near the recessive common-mode level when idle.

Do not continue if either regulator rail is shorted, if the device becomes hot, or if the VIO selector connects both rails.

5.4 Controller Configuration

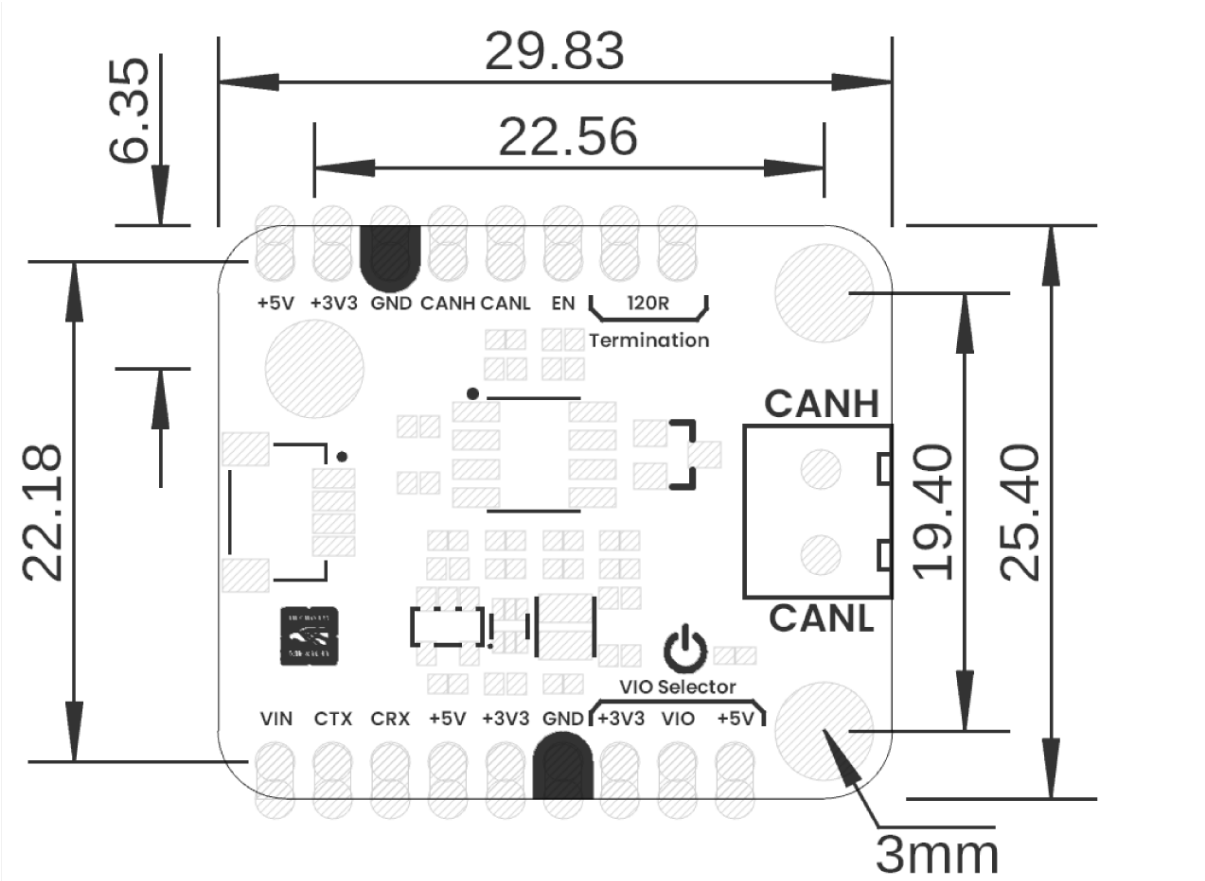
Configure the host CAN controller with the same nominal bit rate as every other node. CAN FD data-phase timing must also match when CAN FD is used. The TCAN1051HVD supports Classic CAN and CAN FD data rates up to 2 Mbps; protocol timing, message identifiers, filters, and acknowledgments are handled by the host controller and software.

5.5 Troubleshooting

Symptom	Checks
No power LED	VIN polarity/level, GND continuity, EN state
Host receives nothing	VIO setting, CRX pin mapping, bit rate, CANH/CANL polarity
Bus errors or missing ACK	Second active node, matching bit rate, two end terminators
Communication degrades with cable length	Twisted pair, stub length, topology, termination, bit rate
Bus held dominant	CTX stuck LOW, incorrect host pin mode, wiring short

6 Mechanical Information

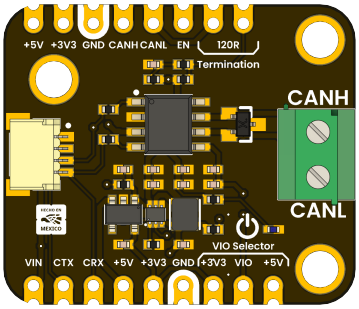
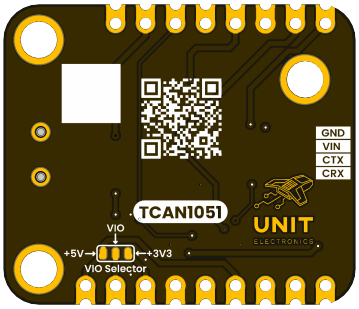
The nominal board outline is **29.83 mm × 25.40 mm**. The drawing below also identifies the 3 mm mounting holes and principal spacing dimensions. All values are in millimeters.



Board Dimensions in Millimeters

Use the released mechanical drawing or a physical sample when designing mating enclosures or production fixtures; the rendered image is not a 1:1 fabrication template.

6.1 Board Views

Top	Bottom
	

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Reference	Link
Product repository	UNIT-Electronics-MX/ unit_devlab_tcan1051hvd_can_transceiver_module
Arduino library	UNIT-Electronics-MX/unit_devlab_tcan1051hvd_library
Product Wiki	UNIT Electronics Wiki — UE0085
TCAN1051HV datasheet	Texas Instruments SLLSES8E
ISO 11898-2 overview	Refer to ISO 11898-2:2016 for the high-speed CAN physical layer

The released schematic, manufacturer datasheets, and hardware resources are available in the product repository under `hardware/` and `hardware/resources/datasheet/`.

The UNIT Electronics Wiki provides additional product information and integration references.

9 Appendix

9.1 Schematic

The complete revision 1 schematic is distributed with this product reference:

Open the UE0085 TCAN1051HVD module schematic

9.2 Design Notes

- R1 provides the switchable 120 Ω termination.
- R2 and R3 form the 2 × 60 Ω split termination.
- C5 (4.7 nF) AC-couples the split-termination midpoint to ground.
- R4 and R5 (22 Ω) are in series with CTX and CRX.
- D1 protects CANH and CANL.
- The S (silent-mode) pin is fixed for normal transceiver operation on this module and is not exposed as a user control.